



Islamic University of Technology
Department of Computer Science and Engineering

Lab 1: Sorting

CSE 4404: Algorithms Lab
Summer 2024-25

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Task A. Triple Sort

Time Limit: 5 minutes | Memory Limit: 512 MB

Given an array a of integers, you need to sort and print it in three different ways:

1. Ascending order.
2. Descending order.
3. Ascending order based on absolute values (if two numbers have the same absolute value, the negative one should come first).

Input Format

The first line of the input contains an integer n ($1 \leq n \leq 100000$) — the size of the array.

The second line contains n space-separated integers $a_1, a_2, \dots, a_n$ ($-100 \leq a_i \leq 100$) — the elements of the array.

Output Format

Output three lines:

- First line: the array sorted in ascending order.
- Second line: the array sorted in descending order.
- Third line: the array sorted in ascending order of absolute values.

Examples

Sample Input	Sample Output
5 1 -3 5 0 -2	-3 -2 0 1 5 5 1 0 -2 -3 0 1 -2 -3 5

Notes

For this task, make 5 submissions with 5 different sorting algorithms:

1. Selection Sort
2. Bubble Sort
3. Insertion Sort
4. Merge Sort
5. Library Sort (C++ STL)

Afterwards, run the local judge to verify the correctness of the solution. Test 1 has an array of 10000 elements, Test 2 has 20000 elements, ... up to Test 10, which has 100000 elements.

Record the runtime for each test case in an Excel file and plot the numbers to generate a graph.

Marks Distribution

Everyone gets 100% marks for this lab.

Practice Problems

Problem	Links
Game	Codeforces: https://codeforces.com/contest/984/problem/A Vjudge: https://vjudge.net/problem/CodeForces-984A
Gravity Flip	Codeforces: https://codeforces.com/contest/405/problem/A Vjudge: https://vjudge.net/problem/CodeForces-405A
Valid Anagram	LeetCode: https://leetcode.com/problems/valid-anagram/
Jagged Swaps	Codeforces: https://codeforces.com/contest/1896/problem/A Vjudge: https://vjudge.net/problem/CodeForces-1896A
Unforgivable Curse (Easy)	Codeforces: https://codeforces.com/contest/1800/problem/E1 Vjudge: https://vjudge.net/problem/CodeForces-1800E1
Unforgivable Curse (Hard)	Codeforces: https://codeforces.com/contest/1800/problem/E2 Vjudge: https://vjudge.net/problem/CodeForces-1800E2